

AYUSH KUMAR SAHU

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Electrical Engineering (CGPA - 8.73)

November 2022 – Present

Rourkela, Odisha

Doon International School, Bhubaneswar

AISCCE – CBSE, Science (PCM) (Percentage - 94.8%)

May 2022

Bhubaneswar, Odisha

Loyola School, Bhubaneswar

CICSE – ICSE (Percentage - 95.6%)

May 2019

Bhubaneswar, Odisha

Relevant Coursework

- Digital Electronics
- Analog Electronics
- Digital IC Design
- Computer Architecture
- Embedded Systems - 8051, 8085
- Signals and Systems

Work Experience

SMSS Laboratory, IIT Kanpur

May 2024 – July 2024

Robotics Intern - Self Navigating Autonomous Rover (Sponsored by Powergrid Corp. of India)

Kanpur, Uttar Pradesh

- Developed an **autonomous and self-navigating** rover for detecting faults in a grid station.
- Implemented **UART protocol** and **PySerial** for serial communication between Arduino and Raspberry Pi 4.
- Built a **ROS-based architecture** on Raspberry Pi 4 to interface with LIDAR, applying **SLAM** and **Nav2** for autonomous navigation and mapping.

Projects

5-Stage Pipelined RISC-V Processor Design

Sept 2025 – Present

Verilog, Xilinx Vivado, GTKWave, RISC-V ISA



- Implemented a **RV32I 5-stage pipelined CPU** (IF, ID, EX, MEM, WB) with full datapath in Verilog.
- Designed **ALU, register file, control unit, immediate generator, hazard detection** and **forwarding logic**.
- Constructed instruction and data memory modules with program-loading support using **\$readmemh**.
- Created a **testbench** featuring instruction preload, data initialization and waveform logging.

Dual-Port SRAM Implementation and Verification

Oct 2025 – Nov 2025

Verilog, System Verilog, Xilinx Vivado, GTKWave



- Built a dual-port synchronous SRAM with **independent read/write operations** on both ports (Port A/B).
- Planned an **arbitration logic** ensuring "Port A wins" in simultaneous writes, validated over 800+ randomized cycles.
- Enabled optional VCD waveform dumping and used **GTKWave** for timing, memory, and read-latency analysis.

Queue Base Architecture for Laundry Management in Hostels

Oct 2024 – Dec 2024

8051 Microcontroller, Assembly Language, RTOS, Proteus



- Developed a **8051 RTOS queue system** in **assembly**, managing RAM-based user data and FIFO logic.
- Designed LCD-keyboard interface; enabled multi-user access via **UART + multiplexers**.
- Achieved **97.5%+ reliability** across **200+ tests**, robust to resets, empty queues, and alerts.

BCD n-bit Shift Multiplier

May 2024

Xilinx Vivado, Verilog



- Designed an **n-bit shift-add multiplier** in Verilog for efficient binary multiplication on FPGA.
- Converted the output to BCD format using the **double-dabble algorithm** for accurate display representation.
- Simulated and verified design in **Xilinx Vivado 2025.1**, ensuring correct functionality and timing on FPGA hardware.

Achievements/Certifications

- Best Innovation Award** – Awarded for making robust obstacle avoidance system for E-HPV by ASME India. 🏆
- Best Paper Award** – For novel research on BMS systems in lithium-ion cells by IEEE India Council. 🏆
- National Finalist** – Positioned 4th nationally against 1500+ teams in Mathworks Minidrone Competition. 🏆
- Machine Learning Specialization** – by Stanford University and DeepLearning.AI. 🏆

Technical Skills

- Languages:** C, C++, Python, Verilog, System Verilog, 8051 & 8085 assembly, Bash Scripting
- Software:** MATLAB, Proteus-8, Keil, NI Multisim, LTSpice, Icarious, Xilinx-Vivado, EDA Playground
- Embedded Tools:** Arduino, Raspberry Pi, ESP-32, Node-MCU 8266, Intel 8085, 8051

Extracurricular Activities

Axiom

June 2023 – June 2025

Chief Event Coordinator & Treasurer

NITR Maths Club

- Lead 20 volunteers in handling "Synapse-2023", annual mathematics competition with 450 students from Rourkela.